Ludovico Grabau

ML Engineer / Medical AI Researcher / Computer Scientist

Engineer and researcher working where machine learning, neuroscience and software meet. I turn research ideas into reliable systems people can use, from brain-computer interfaces and medical language models to cloud and low-level software, I am comfortable in both research labs and engineering teams.

CONTACT ME

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EDUCATION

MSc Artificial Intelligence in Medicine

University of Bern / 2024 - 2026

Deep learning, computer vision, NLP and LLMs, reinforcement learning, medical imaging, neurotechnology.

BSc Computer Science & Communication Systems

HEPIA (HES-SO) Geneva / 2020 - 2023

Software engineering. Thesis on AI-driven narrative characters.

SKILLS

ML & AI

PyTorch TensorFlow/Keras scikit-learn Hugging Face LoRA Optuna RL spaCy LLM APIs

NEURO & SIGNALS

EEG/BCI MNE-Python Real-time DSP Wearables Time-series decoding

PROGRAMMING

Python C/C++ Java TypeScript C# Kotlin Scala Julia SQL R MATLAB Bash

SYSTEMS & CLOUD

Linux Git/GitLab Docker Kubernetes Terraform AWS GCP Azure SLURM CUDA

LANGUAGES

French	●●●●●	Native
English	●●●●●	Fluent
Italian	●●●●●	Fluent
German	●●○○○	B1
Spanish	●○○○○	A1

EXPERIENCE

Master Thesis / Research Assistant

Mar - Sep 2026 / Bern

Neuro-rehaLab & ARTORG Center, University of Bern

- Designed a fair and applicable benchmark of EEG decoders, from classical methods to large foundation models, to read a patient's intended walking direction.
- Showed how a purpose-built model beats foundation models on accuracy, reliability and cost, and added a confidence gate that ignores uncertain predictions.
- Built the real-time pipeline from a dry-EEG headset to gait-robot commands and handed it over as a documented, reproducible system.

Neuroengineer Intern

Jul - Aug 2025 / Biel

Neuro-rehaLab, Berner Fachhochschule (BFH)

- Wrote a Python interface for an EEG headset's C library for live brain-signal streaming.
- Ported the lab's MATLAB workflow to open-source MNE-Python (filters, brain-state analysis, live plots). Becoming the base of my thesis system.

Research Scientist

Feb - Aug 2024 / Geneva

El-Boustani's Laboratory for Cognitive Neurobiology, University of Geneva

- Implemented a mathematical model of motivation and effort in mice as reinforcement-learning simulations, turning equations into working code.
- Searched the simulations backwards for the training conditions that give the best behaviour, handing the wet lab a tested setup for live experiments.

R&D Intern

Jun 2019 - Sep 2020 / Geneva

COSMICS S.A.

- Developed and tested ML models for stock and forex markets and designed how AI trading apps are deployed: my first time building ML inside an engineering team.

Programming Instructor

Summers 2018 - 2022 / Geneva

TechSpark Academy

- Evolved from student to lead instructor teaching Python and game design to children and teens. Learned to explain technical ideas simply and accommodate preferences.

SELECTED PROJECTS

Clinical NLP

Built clinical entity recognition and compared it with fine-tuned BERT and GPT.

Explainable image captioning

Captioned images with a vision transformer and showed where it looks.

Reinforcement learning

Trained DQN and PPO agents; reward shaping made a hard task reliably solvable.

Operating-system kernel

Wrote paging, interrupts, drivers and system calls for an x86 kernel in C.

Cloud and DevOps

Deployed apps with Kubernetes, Terraform and serverless on AWS, GCP and Azure.

High-performance computing

Parallelised image processing with MPI on a SLURM cluster and measured scaling.